Assignment 4

1. **(30 points) MNIST Classification**. The MNIST dataset is a famous dataset in computer vision, which is available at `https://www.kaggle.com/datasets/oddrationale/mnist-in-csv`. The data files mnist_train.csv and mnist_test.csv contain gray-scale images of hand-drawn digits, from 0 through 9.

   Each image is 28 pixels in height and 28 pixels in width, for a total of 784 pixels in total. Each pixel has a single pixel-value associated with it, indicating the lightness or darkness of that pixel, with higher numbers meaning darker. This pixel-value is an integer between 0 and 255, inclusive.

   We work on a 10-category classification problem. The performance measure is the misclassification error on test data. Test data cannot be used for training. The ranking based on the misclassification error is provided at `https://paperswithcode.com/sota/image-classification-on-mnist`.

   - There is no restriction on the method, but your algorithm must be reproducible. You need submit your code for reproducibility. You need to plot the test error against the training epoch.
   - The test errors from all students will be sorted.
     - If your test error is less than 0.62%, you obtain at least 27 points for this problem.
     - If your test error is more than 1.84%, you obtain at most 15 points for this problem.
     - Otherwise, we will assign your score for this problem depending on your ranking.

2. **(30 points) Generative Models for Gaussian Mixtures**. A Gaussian mixture model is a family of distributions whose probability density function is in the following

form:

$$f(x) := \sum_{k=1}^{K} p_k \, \mathcal{N}(x|\mu_{\mathbf{k}}, \Sigma_k),$$

where $\mathcal{N}(\cdot|\mu, \Sigma)$ denotes the Gaussian pdf with mean μ and covariance Σ, and $\{p_1, \ldots, p_K\}$ are mixture weights satisfying

$$\sum_{k=1}^{K} p_k = 1, \quad p_k \geq 0, \quad k \in \{1, \ldots, K\}.$$

Consider a mixture of 8 Gaussians with means $\{(2 \cdot \cos \frac{2\pi k}{8}, 2 \cdot \sin \frac{2\pi k}{8})|k = 0, \ldots 7\}$ and standard deviation 0.02. So $\mu_{\mathbf{k}} = (2 \cdot \cos \frac{2\pi k}{8}, 2 \cdot \sin \frac{2\pi k}{8}) \in \mathbb{R}^2$ and $\Sigma_k = 0.02^2 I_2 \in \mathbb{R}^{2\times 2}$ for $k = 0, \ldots, 7$. The random samples are shown in Figure 1.

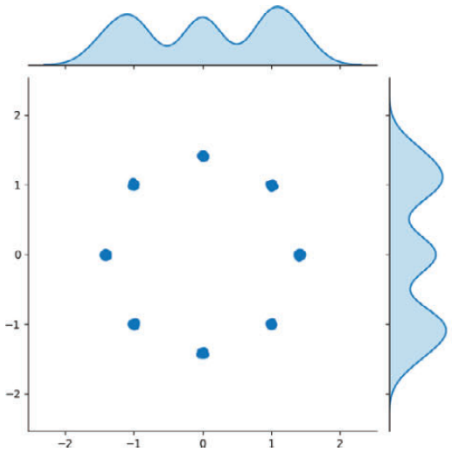


Figure 1: Gaussian mixtures

- Simulate 20,000 samples from the Gaussian mixture distribution. You should use the normal information for sampling, so the resulted samples follow exactly the Gaussian mixture distribution. These will be our training data.

- Implement a vanilla GAN model to obtain the discriminator and generator. After training, obtain 1600 samples from your GAN model, which should look like the samples from the Gaussian mixture in Figure 1.
- How good are these results? Since we know the true underlying distribution, we can provide a method to quantify your samples. Maximum mean discrepancy (MMD) is a kernel based statistical test used to determine whether given two distribution are the same. The code is availabe at `https://www.kaggle.com/code/onurtunali/maximum-mean-discrepancy/notebook`. Report the MMD between your generated samples and real samples. You should generate another set of 1600 real samples by generating 200 samples from each bivariate normal distribution.